

Scalability & Future Plan

Date	02 July 2026
Team ID	CCAP-01
Project Name	Credit Card Approval Prediction
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Small, sample-sized dataset (application_record.csv and credit_record.csv)	Model may not generalize well to a real-world, more diverse applicant population	High
2	Prediction history stored in a local SQLite file (prediction_history.db)	Not suitable for concurrent multi-user access or high write throughput	Medium
3	Model retraining is a manual step (running train.py)	No automated pipeline to retrain or refresh the model as new data arrives	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Flask development server (python run.py), a single-process instance intended for local/demo use	Serve with Gunicorn behind Nginx, containerize with the included Dockerfile, and scale horizontally with a load balancer or container orchestration (e.g. Kubernetes)
2	Data Storage	Static CSV files for training data and a local SQLite database for prediction history	Move datasets and prediction history to a managed relational database (e.g. PostgreSQL) or cloud data warehouse to support larger volumes and concurrent access

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
3	Performance	Single saved sklearn pipeline (best_model.pkl) loaded and used per request	Cache the loaded model as a singleton, batch inference requests where possible, and consider a dedicated model-serving layer (e.g. FastAPI microservice or MLflow) for lower latency at higher traffic
4	Security	Basic signup/login flow with a Flask SECRET_KEY managed through .env	Enforce HTTPS, strengthen password hashing and session handling, add rate limiting, and extend the existing Google login configuration into full OAuth/SSO support

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Automated retraining pipeline with model monitoring and drift detection	Q3 2026	Keeps the model current as new applicant data arrives and flags performance degradation early
Phase 3	Migrate to a cloud-hosted database (PostgreSQL) and containerized deployment using the existing Dockerfile and Procfile	Q4 2026	Supports multiple concurrent users and higher request throughput
Phase 4	Fairness and bias auditing, a larger validated dataset, and a compliance review	Q1 2027	Moves the project toward production readiness for real credit-decision use, addressing the limitation noted in the project README